

Table 1: Comparative Analysis of State-of-the-Art Deep Learning Models for Surgical Instrument Recognition

Ref	Task	Data	Edu	Models					Metrics					
				RTD	T&S	SB	FM	HC	Perf.	Over.	Err.	Flow	Spd.	Oth.
[1]	Detect / Track		✓	✓					✓			✓		
[2]	Detect			✓					✓			✓		
[3]	Detect	✓		✓					✓			✓		✓
[4]	Detect	✓		✓					✓			✓		
[5]	SWR		✓		✓				✓					
[6]	SWR	✓				✓		✓	✓			✓		
[7]	Detect						✓		✓			✓		
[8]	Segment					✓		✓		✓				
[9]	Segment					✓		✓	✓	✓				
[10]	Segment	✓				✓	✓	✓		✓				
[11]	Detect			✓		✓		✓	✓			✓		
[12]	SWR		✓		✓	✓		✓						✓
[13]	Segment	✓		✓		✓		✓		✓				
[14]	Detect / Track	✓		✓	✓			✓	✓			✓	✓	
[15]	Pose	✓			✓			✓			✓			
[16]	Track		✓		✓			✓			✓			✓
[17]	SWR			✓				✓	✓				✓	
[18]	Segment	✓				✓		✓		✓			✓	
[19]	Pose	✓						✓			✓		✓	
[20]	Pose / Track	✓				✓		✓			✓		✓	
[21]	Pose / Segment		✓		✓			✓		✓	✓		✓	
[22]	Detect / Pose	✓		✓				✓			✓	✓	✓	
[23]	Pose / Track				✓	✓		✓			✓	✓	✓	
[24]	Segment					✓		✓		✓	✓		✓	

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Ref	Task	Data	Edu	Models					Metrics					
				RTD	T&S	SB	FM	HC	Perf.	Over.	Err.	Flow	Spd.	Oth.
[25]	Detect / Pose		✓			✓		✓		✓		✓	✓	
[26]	Segment / Track	✓		✓		✓		✓		✓		✓	✓	
[27]	Track / Pose		✓	✓				✓			✓	✓	✓	
[28]	SWR / Detect	✓		✓	✓			✓	✓	✓		✓	✓	
[29]	Detect	✓		✓	✓	✓		✓				✓	✓	✓
[30]	Segment	✓			✓	✓		✓		✓	✓		✓	
[31]	Track	✓		✓	✓	✓		✓					✓	✓
[32]	Pose	✓		✓	✓	✓	✓	✓					✓	✓
[33]	Manage/Classify	✓		✓	✓	✓		✓	✓			✓		✓
[34]	Segment	✓		✓		✓		✓	✓	✓		✓		
[35]	Pose	✓		✓		✓		✓			✓		✓	✓
[36]	Detect/Classify	✓		✓		✓		✓	✓			✓	✓	✓
[37]	Detect	✓		✓		✓		✓	✓			✓		✓
[38]	Classify/SRW/Det	✓			✓	✓	✓	✓				✓	✓	✓
[39]	Seg/Track/Det	✓		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
[40]	Pose / Track		✓	✓			✓			✓	✓	✓	✓	
[41]	Segment / Track	✓		✓	✓	✓		✓	✓	✓		✓	✓	✓
[42]	Pose	✓		✓		✓		✓	✓	✓	✓		✓	✓
[43]	Segment	✓				✓	✓	✓		✓	✓			✓
[44]	Track	✓				✓		✓		✓	✓		✓	✓
[45]	Segment	✓		✓		✓		✓	✓	✓	✓		✓	✓
[46]	Segment	✓				✓		✓		✓	✓		✓	✓
[47]	Detect / Segment	✓	✓	✓		✓		✓	✓	✓	✓			✓
[48]	Segment	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓		✓

Abbreviations: **T&S:** Temporal & Sequential; **SB:** Segmentation Backbones; **FM:** Foundation Models; **HC:** Hybrid / Custom; **Over.:** Overlap; **Err.:** Error; **Spd.:** Speed; **Oth.:** Others; **SWR:** Surgical Workflow Recognition; **Edu:** Education/Simulation.